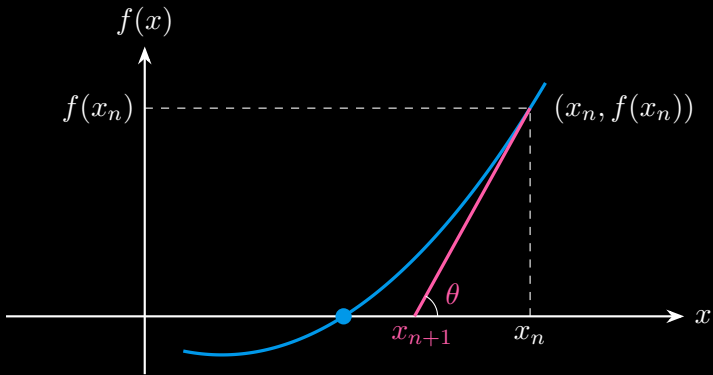


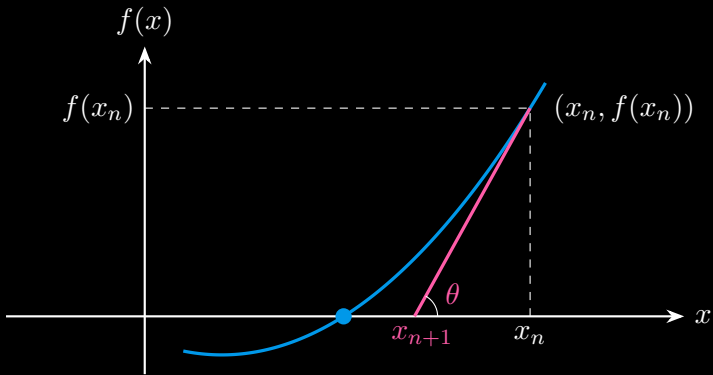
Intuitive Understanding of Newton's Method

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{0 - f(x_n)}{x_{n+1} - x_n} = f'(x_n)$$



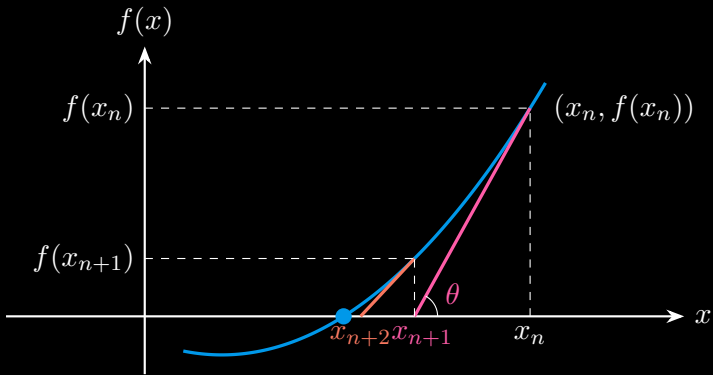
Intuitive Understanding of Newton's Method

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \quad \Rightarrow \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



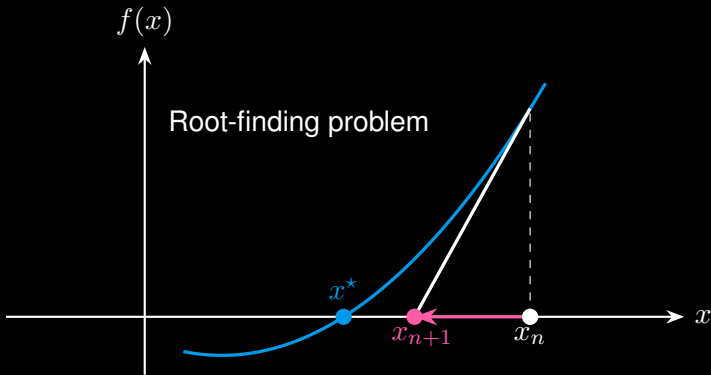
Intuitive Understanding of Newton's Method

$$x_{n+2} = x_{n+1} - \frac{f(x_{n+1})}{f'(x_{n+1})}$$



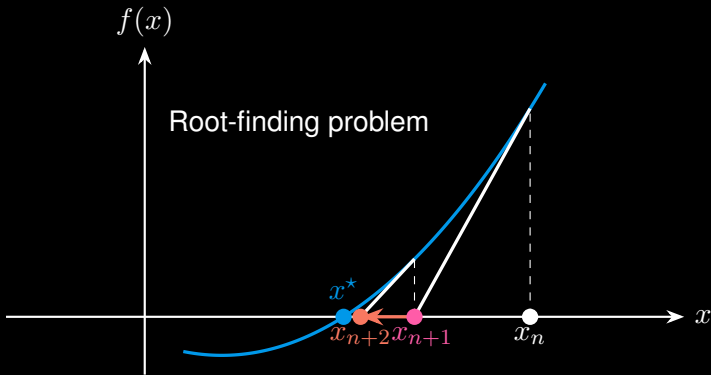
Intuitive Understanding of Newton's Method

$$x := x - \frac{f(x)}{f'(x)} \quad (\text{update } x \text{ until convergence})$$



Intuitive Understanding of Newton's Method

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Unconstrained Nonlinear Optimization

Optimization problem:

$$\min_x \underbrace{f(x)}_{\text{twice-differentiable}} \quad \Rightarrow \quad \text{Find } f'(x) = 0$$

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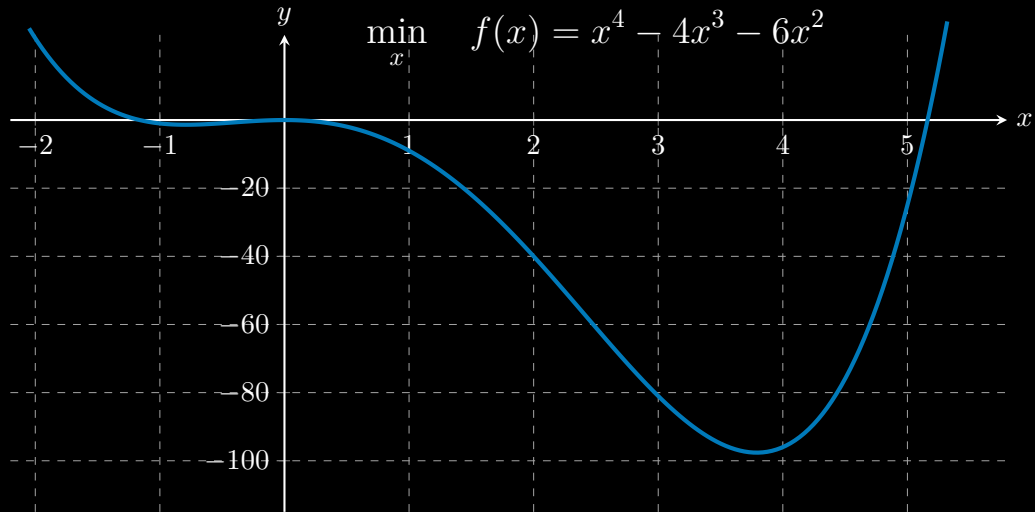
Newton's method:

$$x := x - \frac{f'(x)}{f''(x)} \quad (\text{update } x \text{ until convergence})$$

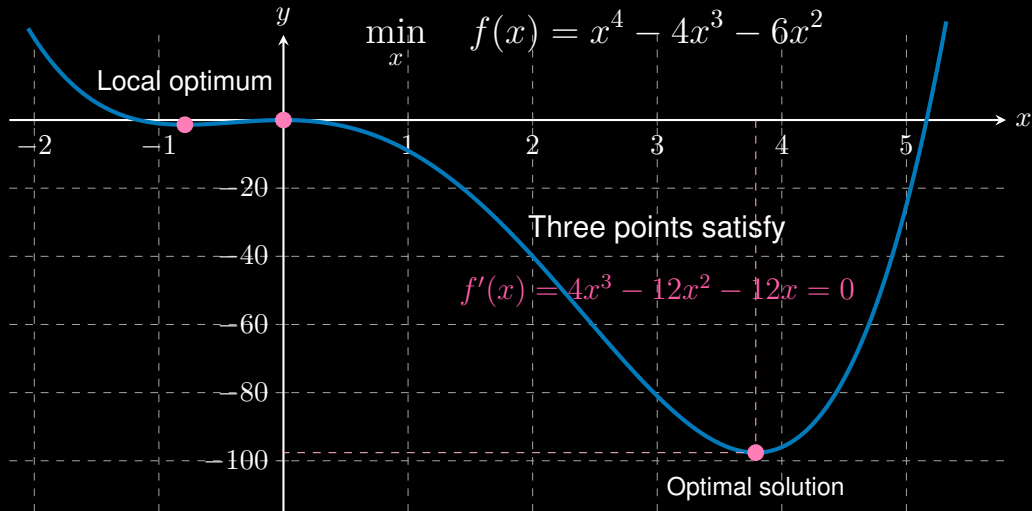
with

$$\begin{array}{ll} f'(x) & \text{first-order derivative} \\ f''(x) & \text{second-order derivative} \end{array}$$

Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization

Optimization problem:

$$\min_x f(x) = x^4 - 4x^3 - 6x^2$$

Newton's method:

$$x := x - \frac{f'(x)}{f''(x)} \quad (\text{update } x \text{ until convergence})$$

with

$$f'(x) = 4x^3 - 12x^2 - 12x \quad \text{and} \quad f''(x) = 12x^2 - 24x - 12$$

Unconstrained Nonlinear Optimization

Optimization problem:

$$\min_x f(x) = x^4 - 4x^3 - 6x^2$$

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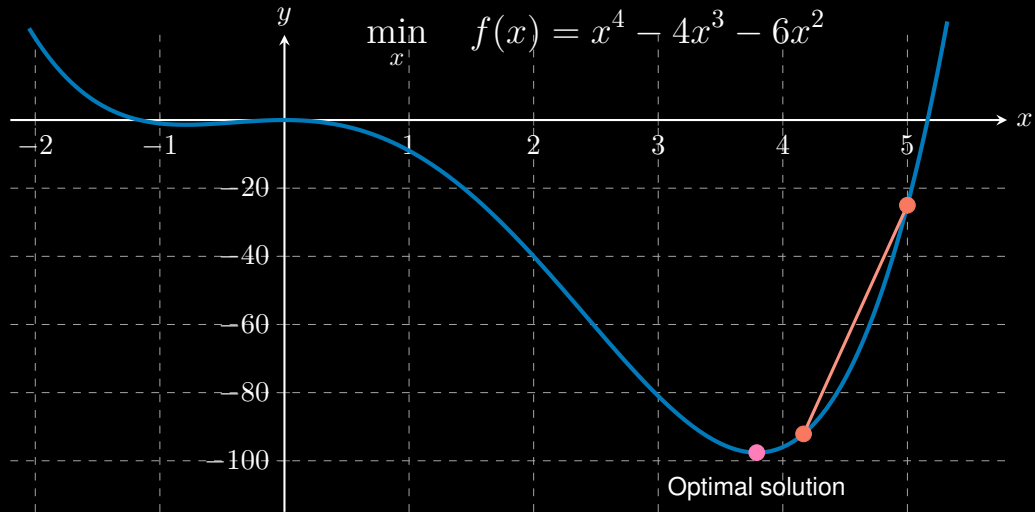
with

$$f'(x) = 4x^3 - 12x^2 - 12x \quad \text{and} \quad f''(x) = 12x^2 - 24x - 12$$

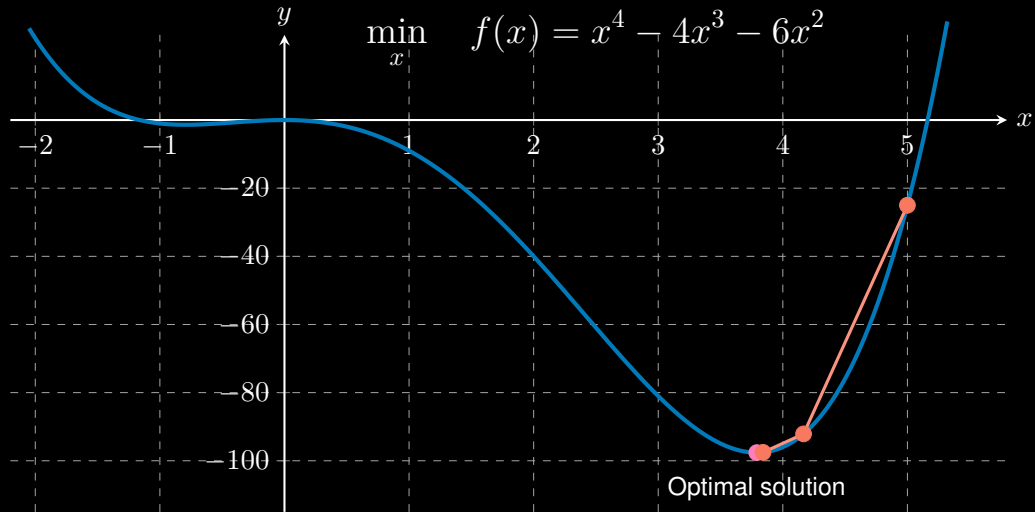
Given a starting point $x_0 = 5$:

$$f'(5) = 140, \quad f''(5) = 168 \quad \Rightarrow \quad x := 5 - \frac{f'(5)}{f''(5)} = 4.1667$$

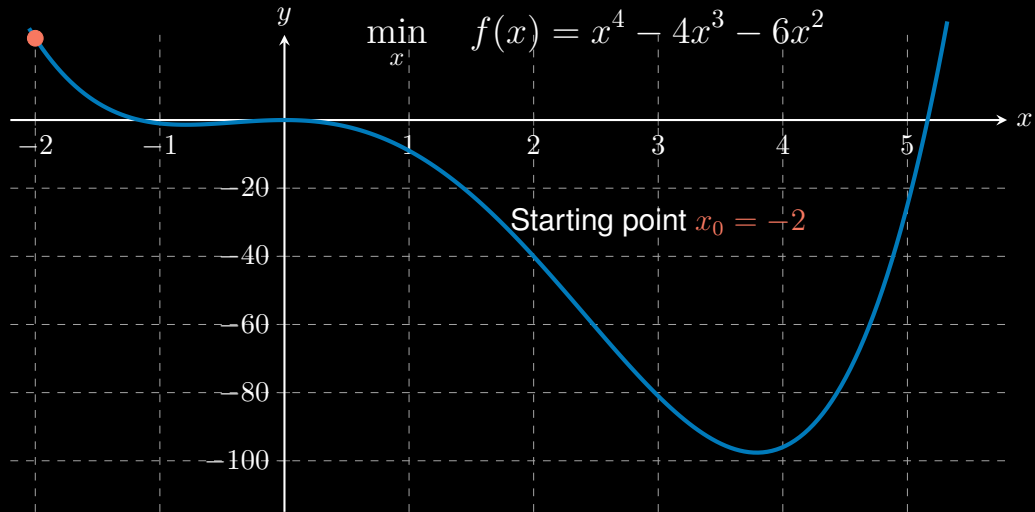
Unconstrained Nonlinear Optimization



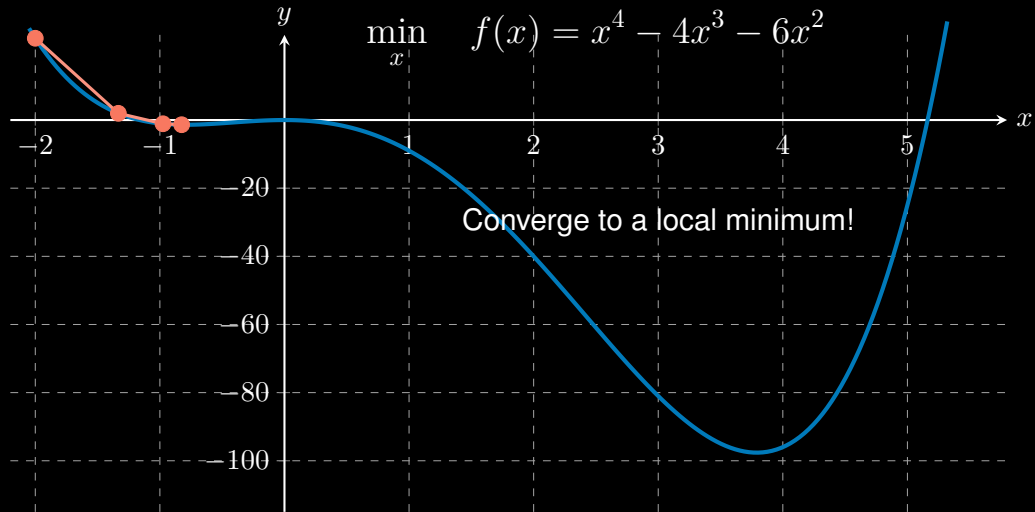
Unconstrained Nonlinear Optimization



Unconstrained Nonlinear Optimization

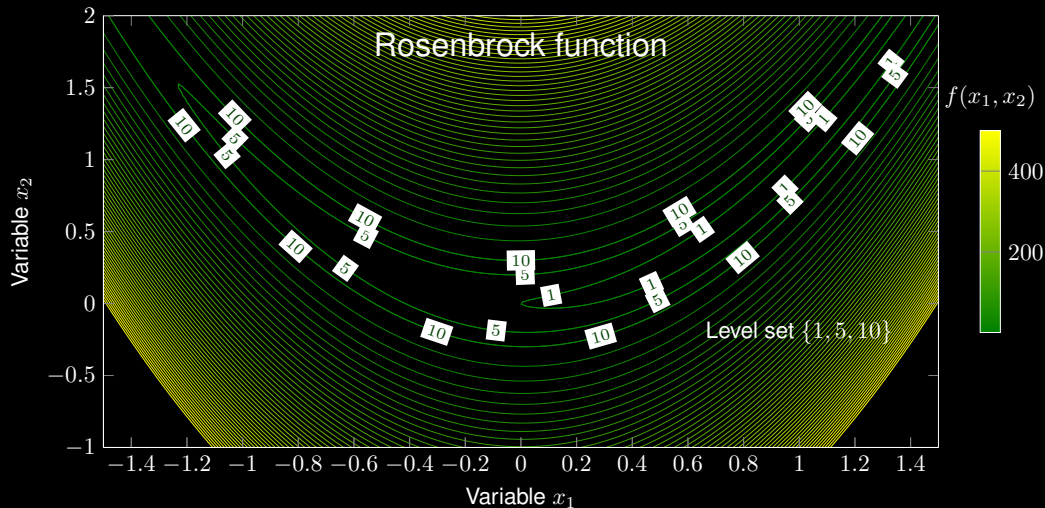


Unconstrained Nonlinear Optimization



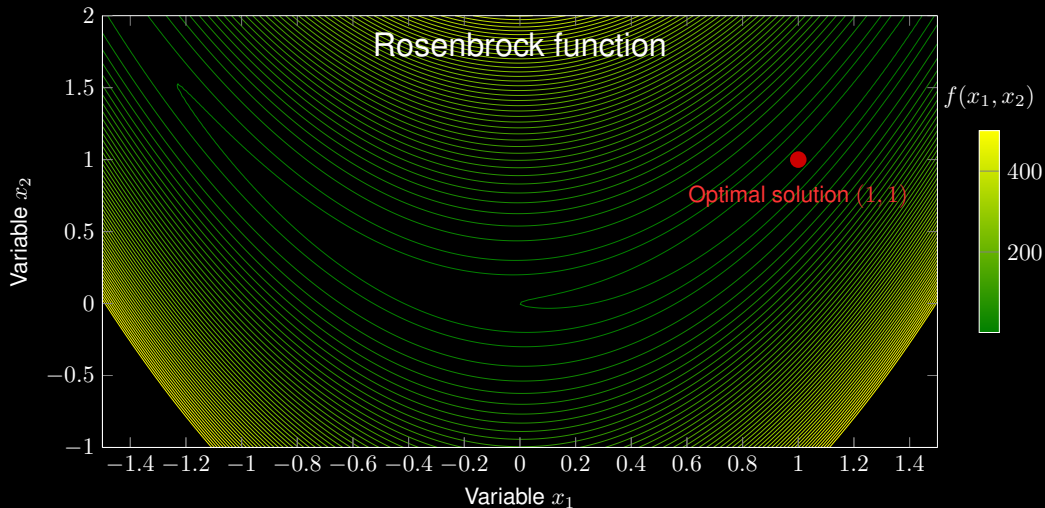
Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$



Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$



First- and Second-Order Derivatives

$$\min_{x_1, x_2} \underbrace{f(x_1, x_2)}_{\text{twice-differentiable}} = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$

Unconstrained Nonlinear Optimization

The optimization can be converted into a root finding problem:

$$\min_{x_1, x_2} f(x_1, x_2) \quad \Leftrightarrow \quad \text{Find} \quad \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \mathbf{0} \quad (\text{gradient} = \mathbf{0})$$

Newton's method for root finding:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \mathbf{H}^{-1} \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}$$

with

$$\mathbf{H} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} \quad (\text{Hessian matrix})$$

Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$

